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Misspecified proportional hazard models

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SUMMARY

The properties of an estimator based on a proportional hazards model are investigated when the model is incorrect. The estimator from the partial likelihood is shown to be consistent for a parameter that is defined implicitly. The results are used to investigate the effects on estimation if the true model is accelerated failure time, or if covariates are omitted from the proportional hazards model.

Some key words: Accelerated failure time; Estimation; Failure time model; Model misspecification; Partial likelihood; Proportional hazards.

1. INTRODUCTION

The proportional hazards model for the time to failure, T, specifies that the covariates $Z^{\mathrm{T}} = (Z_1, \ldots, Z_p)$ act multiplicatively on the failure rate or hazard. In this paper, the hazard function given Z is taken to be

$$\lambda(t; Z) = \lambda_0(t) \exp(\beta^{\mathrm{T}} Z), \tag{1·1}$$

where $\beta^{\mathrm{T}} = (\beta_1, \ldots, \beta_p)$ is a vector of regression parameters, and $\lambda_0(t)$ is an arbitrary baseline hazard function. Estimation of β can be based on the partial likelihood (Cox, 1972, 1975)

$$L(\beta) = \prod_{i=1}^{n} \left\{ \exp(\beta^{\mathrm{T}} Z_i) \Big/ \sum_{j \in R(t_i)} \exp(\beta^{\mathrm{T}} Z_j) \right\}^{\delta_i}, \tag{1·2}$$

where $R(t)$ is the set of labels attached to individuals at risk at time $t-$, t_i is the observation time of item i, and $\delta_i = 1$ if t_i is a failure and $\delta_i = 0$ otherwise.

An alternative class of models for T is the accelerated failure time model

$$\log T = \alpha_0 + \alpha^{\mathrm{T}} Z + \sigma W, \tag{1·3}$$

where $\alpha^{\mathrm{T}} = (\alpha_1, \ldots, \alpha_p)$ and $\sigma > 0$ are unknown parameters, and W has density $f(w)$ and distribution function $F(w)$. In the special case in which W has the extreme value distribution for a minimum, it is clear that the partial likelihood (1·2) estimates the parameter $\beta = \sigma^{-1}\alpha$.

In this paper, we examine the properties of estimators from (1·2) in a more general context. The results are used to investigate the effects on estimation if the true model is accelerated failure time, or if covariates are omitted from the proportional hazards model. A special case of interest is that of a randomized treatment assignment in which Z_1 specifies the treatment group; we examine the effect on estimation of β_1 when the covariates $Z_2, \ldots, Z_p$ are omitted from the model.

Solomon (1984, 1986) considers some aspects of the same problem, and her results overlap those in an unpublished University of Waterloo thesis by C. A. Struthers. The present work indicates how this problem may be approached within the framework of counting processes and establishes some consistency results. The relationship between estimation and local asymptotic efficiency is also discussed.

2. NOTATION AND GENERAL RESULTS

Following Andersen & Gill (1982), let $(N_1, \ldots, N_n)$ for $n = 1, 2, \ldots$ be a multivariate counting process in which $N_i(t)$ records the number of failures, 0 or 1, in $[0, t]$ for the ith item. We suppose that $N_i(t)$ is right continuous and that no two components have simultaneous jumps. As usual, the probability model is (Ω, F, P) and $(F_t\colon 0 \leqslant t < \infty)$ is a right continuous, nondecreasing family of sub σ-algebras where F_t comprises events defined in $[0, t]$. Local martingales, predictable processes, etc. are defined with reference to these sub σ-algebras.

Suppose that N_i has random intensity process of the form

$$Y_i(t)\lambda(t; Z_i) \quad (i = 1, \ldots, n), \tag{2·1}$$

where $Y_i(t)$, a predictable process, takes the value 1 if the ith item is at risk at time $t-$ and 0 otherwise, and Z_i is the covariate for the ith item. Here $\lambda(t; Z_i)$ specifies the hazard or failure rate for an item at risk at time t with covariate Z_i. Although this set-up allows for more general censoring schemes (Andersen, Borgan et al., 1982) we assume, for convenience, that (N_i, Y_i, Z_i) $(i = 1, \ldots, n)$ are independent and identically distributed replicates of (N, Y, Z). In this case, the processes

$$M_i(t) = N_i(t) - \int_0^t Y_i(x)\lambda(x; Z_i)\, dx \quad (i = 1, \ldots, n) \tag{2·2}$$

are independent local square integrable martingales on $[0, \infty]$ with covariance process $\langle M_i, M_j\rangle(t) = 0$, for $i \neq j$, and

$$\langle M_i, M_i\rangle(t) = \int_0^t Y_i(x)\lambda(x; Z_i)\, dx \quad (i = 1, \ldots, n).$$

For convenience, it is also assumed that Z_i is a scalar. The results generalize to the vector case in a straightforward manner.

Suppose that it is incorrectly assumed that the proportional hazards model is applicable. Thus, we postulate a random intensity process of the form

$$Y_i(t)\lambda_0(t)\exp(\beta Z_i) \quad (i = 1, \ldots, n). \tag{2·3}$$

It is convenient here to introduce the notation

$$S^{(j)}(x) = n^{-1}\sum_{i=1}^{n} Z_i^j Y_i(x)\lambda(x; Z_i), \quad s^{(j)}(x) = E\{S^{(j)}(x)\},$$

$$S^{(j)}(\beta, x) = n^{-1}\sum_{i=1}^{n} Z_i^j Y_i(x)\exp(\beta Z_i), \quad s^{(j)}(\beta, x) = E\{S^{(j)}(\beta, x)\},$$

for $j = 0, 1, 2$, where $0^0 = 1$, and expectations are taken with respect to the true model (2·1). Under the assumed model (2·3), β is estimated from the partial likelihood (1·2) or from its logarithm, $l(\beta, \infty)$, where

$$l(\beta, t) = \sum_{i=1}^{n} \int_0^t \beta Z_i \, dN_i(x) - \int_0^t \log\{S^{(0)}(\beta, x)\} \sum_{i=1}^{n} dN_i(x).$$

The corresponding estimator $\hat{\beta}$ can be shown to be the unique solution to $\partial l(\beta, \infty)/\partial \beta = U(\beta, \infty) = 0$, where

$$U(\beta, t) = \sum_{i=1}^{n} \int_0^t \left\{ Z_i - \frac{S^{(1)}(\beta, x)}{S^{(0)}(\beta, x)} \right\} dN_i(x). \tag{2·4}$$

THEOREM 2·1. *The maximum partial likelihood estimator $\hat{\beta}$ is a consistent estimator of β^*, where β^* is the solution to the equation $h(\beta) = 0$ with*

$$h(\beta) = \int_0^\infty s^{(1)}(x) \, dx - \int_0^\infty \frac{s^{(1)}(\beta, x)}{s^{(0)}(\beta, x)} s^{(0)}(x) \, dx \tag{2·5}$$

if the following conditions hold.

Condition 1. *There exists a neighbourhood B of β^* such that for each $t < \infty$*

$$\sup_{x \in [0,t], \beta \in B} |S^{(0)}(\beta, x) - s^{(0)}(\beta; x)| \to 0$$

in probability as $n \to \infty$, $s^{(0)}(\beta, x)$ is bounded away from zero on $B \times [0, t]$, and $s^{(0)}(\beta, x)$ and $s^{(1)}(\beta, x)$ are bounded on $B \times [0, t]$.

Condition 2. *For each $t < \infty$, $\int s^{(2)}(x) \, dx < \infty$, where the integral is over $(0, t)$.*

Proof. This proof is similar to the proof of consistency of the maximum partial likelihood estimator given by Andersen & Gill (1982). Using their methods, it can be shown that, under Conditions 1 and 2, $n^{-1} l(\beta, \infty)$ converges in probability to

$$H(\beta) = \beta \int_0^\infty s^{(1)}(x) \, dx - \int_0^\infty \log\{s^{(0)}(\beta, x)\} s^{(0)}(x) \, dx$$

for each $\beta \in B$. By the boundedness conditions in Condition 1 the first derivative of $H(\beta)$ equals $h(\beta)$ and the second derivative is

$$-\int_0^\infty \left[\frac{s^{(2)}(\beta, x)}{s^{(0)}(\beta, x)} - \left\{ \frac{s^{(1)}(\beta, x)}{s^{(0)}(\beta, x)} \right\}^2 \right] s^{(0)}(x) \, dx.$$

The second derivative is always negative and by definition $h(\beta^*) = 0$. Therefore, for each $\beta \in B$, $n^{-1} l(\beta, \infty)$ converges in probability to the concave function $H(\beta)$ which has a unique maximum at $\beta = \beta^*$. Since $\hat{\beta}$ maximizes the concave function $n^{-1} l(\beta, \infty)$ it follows by some convex analysis (Andersen & Gill, 1982, Appendix 2) that $\hat{\beta}$ converges in probability to β^*. □

3. SOME EXAMPLES

3·1. *Sampling from accelerated failure time models*

If the true model is (1·3) and $p = 1$, expression (2·1) takes the form

$$Y_i(t) \lambda_f\{(\log t - \alpha_0 - \alpha Z_i)/\sigma\}/(\sigma t), \tag{3·1}$$

where $f(w)$, $F(w)$ and $\lambda_f(w)=f(w)/\{1-F(w)\}$ are the density, distribution and 'hazard' functions, respectively, of the error variable W. Consider the random censorship model in which $C(t; Z)$ is the survivor function of the censoring time for an individual with covariate Z. If a proportional hazards analysis is used then, by Theorem 2·1, $\hat{\beta}$ is consistent for β^*, where β^* is the solution to the equation

$$0=\int_0^\infty E\left(g(t; Z)C(t; Z)\left[Z-\frac{E\{Ze^{\beta Z}G(t; Z)C(t; Z)\}}{E\{e^{\beta Z}G(t; Z)C(t; Z)\}}\right]\right) dt, \tag{3·2}$$

where $G(t; Z)=\text{pr}\,(T\geq t; Z)$ and

$$g(t; Z)=-dG(t; Z)/dt=f\{(\log t-\alpha_0-\alpha Z)/\sigma\}/(\sigma t).$$

Differentiation of (3·2) yields a first approximation to β^*, namely

$$\beta^*\simeq\left[\alpha\frac{\partial\beta^*}{\partial\alpha}\right]_{\alpha=0}=-\alpha E\left\{\frac{\lambda_f'(W)}{\lambda_f(W)}V(W)\right\}\bigg/[\sigma E\{V(W)\}], \tag{3·3}$$

where

$$V(w)=E\{Z^2C(t; Z)\}-[E\{ZC(t; Z)\}]^2/E\{C(t; Z)\} \tag{3·4}$$

and $w=(\log t-\alpha_0)/\sigma$. When there is no censoring, (3·3) reduces to $\beta^*\simeq-\alpha E\{\lambda_f(W)\}/\sigma$.

This derives, in a different way, results of Solomon (1984, 1986). As she notes, (3·3) illustrates that the use of the proportional hazards model for analysis when the accelerated failure time model is true leaves unchanged, to first order, the relative importance of the covariates. It can be shown that the approximation holds to second order when Z is symmetric and there is no censoring.

In an unpublished technical report, C. A. Struthers has shown that the asymptotic relative efficiency of the log rank test compared to the censored linear rank test with score generating density f is

$$\sigma^2\left(\left[\frac{\partial\beta^*}{\partial\alpha}\right]_{\alpha=0}\right)^2 E_f\{V(W)\}\bigg/E_f\left[\left\{\frac{\lambda_f'(W)}{\lambda_f(W)}\right\}^2 V(W)\right]$$

when sampling is from the accelerated failure time model with error density $f(.)$. This illustrates the relationship between asymptotic relative efficiency and the first-order expression for β^*, (3·3). See also Lagakos & Schoenfeld (1984).

3·2. *Proportional hazards with missing covariates*

Suppose that the true model is proportional hazards with respect to two independent covariates Z_1, Z_2 and that the true random intensity process is of the form

$$Y_i(t)\lambda_1(t)\exp(\alpha^{\mathrm{T}}Z_i), \tag{3·5}$$

where $Z_i^{\mathrm{T}}=(Z_{i1}, Z_{i2})$, $\alpha^{\mathrm{T}}=(\alpha_1, \alpha_2)$ and $\lambda_1(t)$ is a baseline hazard function. Consider again a random censorship model in which the survivor function for the censoring time of an individual with covariate vector $Z^{\mathrm{T}}=(Z_1, Z_2)$ is $C(t; Z_1)$, independent of Z_2. In this section, we consider estimation of α_1 when the model assumed is

$$Y_i(t)\lambda_0(t)\,e^{\beta Z_{1i}}. \tag{3·6}$$

We are here concerned with evaluating the effects on estimation of omitting a covariate from the model.

It is useful to note an analogous problem in normal linear regression in which $Y=\alpha_0+\alpha_1 Z_1+\alpha_2 Z_2+\sigma W$, where Z_1, Z_2 and W are independent. In the analysis of this, the appropriate reference set has Z_1 and Z_2 fixed, and the least squares estimator is unbiased and has conditional variance τ^2 say. If, however, Z_2 is unavailable and omitted from the model, the appropriate reference set fixes Z_1 but allows Z_2 to vary. The least squares estimator of α_1 would, in this reference set, be unbiased and have variance larger than τ^2 at least in large samples. A closer analogy with the proportional hazards model is obtained by considering estimation of β_1 in the model

$$Y=\alpha_0+\beta_1\sigma Z_1+\beta_2\sigma Z_2+\sigma W$$

versus β in the model with Z_2 omitted

$$Y=\alpha_0+\beta\sigma_1 Z_1+\sigma_1 W.$$

If $\beta_1 \neq 0$ and $\beta_2 \neq 0$, then $|\beta|<|\beta_1|$ and the estimator of β in the second model with Z_2 omitted will be asymptotically biased for β_1, in fact biased toward zero, and can be shown to have smaller asymptotic variance than the estimator of β_1 based on the true model with Z_2 included.

From (2·5), the estimator $\hat{\beta}$ that arises from the assumed model (3·6) is consistent for β^*, the solution to

$$0=\int_0^\infty E\left(h(t;Z_1)C(t;Z_1)\left[Z_1-\frac{E\{Z_1 e^{\beta Z_1}H(t;Z_1)C(t;Z_1)\}}{E\{e^{\beta Z_1}H(t;Z_1)C(t;Z_1)\}}\right]\right)dt, \tag{3·7}$$

where

$$h(t;z_1)=-dH(t;Z_1)/dt=\lambda_1(t)E[e^{\alpha^{\mathrm{T}}Z}\exp\{e^{-\alpha^{\mathrm{T}}Z}\Lambda_1(t)\}|Z_1]$$

is the true density of the failure time T given Z_1 and $\Lambda_1(t)=\int\lambda_1(u)\,du$, where the integral is over $(0,t)$.

If $\alpha_1=0$, then both (3·5) and (3·6) are correct models, and $\beta^*=0$ is the solution to (3·7). Further, from results of Andersen & Gill (1982), $n^{\frac{1}{2}}\hat{\beta}$ is asymptotically $N(0,I^{-1})$, where

$$I=\int_0^\infty W(t)h(t;0)\,dt,\quad W(t)=E\{Z_1^2C(t;Z_1)\}-[E\{Z_1C(t;Z_1)\}]^2/E\{C(t;Z_1)\}.$$

If the correct model (3·5) were used, the maximum partial likelihood estimators $(\hat{\alpha}_1,\hat{\alpha}_2)$ are asymptotically independent, and if $\alpha_1=0$ it can be seen that

$$n^{\frac{1}{2}}(\hat{\alpha}_1,\hat{\alpha}_2-\alpha_2)^{\mathrm{T}}\to N\{(0,0)^{\mathrm{T}},\operatorname{diag}(I^{-1},I_2^{-1})\}$$

in distribution where I_2^{-1} is the asymptotic variance of $\hat{\alpha}_2$. Analogous to the normal case, the two estimators $\hat{\beta}$ and $\hat{\alpha}_1$ have the same asymptotic distribution if $\alpha_1=0$. The asymptotic relative efficiency of a test of $\alpha_1=0$ based on $\hat{\beta}$ compared to a test based on $\hat{\alpha}_1$ is the square of

$$\left[\frac{\partial\beta^*}{\partial\alpha_1}\right]_{\alpha_1=0}=I^{-1}\int_0^\infty\psi(t)W(t)g(t;0)\,dt, \tag{3·8}$$

where

$$\psi(t)=-\frac{d}{dw}\log\frac{E\{e^{w+\alpha_2Z_2}\exp(-e^{w+\alpha_2Z_2})\}}{E\{\exp(-e^{w+\alpha_2Z_2})\}}$$

with $w=\log \Lambda_1(t)$. If Z_2 takes on values 0 and 1 each with probability $\frac{1}{2}$ and there is no censoring then the square of (3·8) equals 0·83 for $\alpha_2=\log 2$, 0·69 for $\alpha_2=\log 3$ and 0·61 for $\alpha_2=\log 4$.

For the case $\alpha_1>0$ only model (3·5) is correct; $\alpha_1<0$ is similar.

THEOREM 3·1. *Let β^* be defined by* (3·7). *If $\alpha_1>0$ then $0<\beta^*<\alpha_1$.*

Proof. Define

$$S_1(\beta)=\int_0^\infty E\{Z_1\lambda(t;Z)G(t;Z)\}\,dt-\int_0^\infty \frac{E\{Z_1\, e^{\beta Z_1+\alpha_2 Z_2}G(t;Z)\}}{E\{e^{\beta Z_1+\alpha_2 Z_2}G(t;Z)\}}\,E\{\lambda(t;Z)G(t;Z)\}\,dt,$$

where

$$G(t;Z)=C(t;Z_1)\exp\{-e^{\alpha^{\mathrm{T}}Z}\Lambda_1(t)\}$$

and $\lambda(t;Z)=\exp(\alpha^{\mathrm{T}}Z)\lambda_1(t)$. Since $S_1(\alpha_1)=0$ and $dS_1(\beta)/d\beta<0$ for all β, therefore $S_1(\beta)$ is a monotone decreasing function of β which crosses the x axis at $\beta=\alpha_1$. It can be shown that $S_1(\beta^*)>0$ and therefore $\beta^*<\alpha_1$.

Similarly, if we denote by $S(\beta)$ the right-hand side of (3·7), then $S(\beta)$ is monotone decreasing in β with $S(0)>0$ and $S(\beta^*)=0$. It follows that $\beta^*>0$. □

To examine the magnitude of this bias consider the following example. Suppose there is no censoring and $\mathrm{pr}\{Z_1=i, Z_2=j\}=\frac{1}{4}$ for $i,j=0,1$. In this case β^* is the solution to the equation $S(\beta;\alpha_1,\alpha_2)=0$, where

$$\begin{aligned}S(\beta;\alpha_1,\alpha_2)=\tfrac{1}{2}-\int_0^\infty &(e^{\beta}\{\exp(-t\,e^{\alpha_1})+\exp(-t\,e^{\alpha_1+\alpha_2})\}\\ &\times\{e^{-t}+\exp(\alpha_2-t\,e^{\alpha_2})+\exp(\alpha_1-t\,e^{\alpha_1})+\exp(\alpha_1+\alpha_2-t\,e^{\alpha_1+\alpha_2})\}\\ &\times[e^{-t}+\exp(-t\,e^{\alpha_1})+e^{\beta}\{\exp(-t\,e^{\alpha_1})+\exp(-t\,e^{\alpha_1+\alpha_2})\}]^{-1})\,dt.\end{aligned}$$

Note that if β^* is the solution to the equation $S(\beta;\alpha_1,\alpha_2)=0$, then β^* is also the solution to $S(\beta;\alpha_1,-\alpha_2)=0$ and $-\beta^*$ is the solution to $S(\beta;-\alpha_1,\alpha_2)=0$ and $S(\beta;-\alpha_1,-\alpha_2)=0$. Table 1 gives values of β^* for selected values of α_1 and α_2. Also included in the table are values of e^{α_1} and e^{β^*} since e^{α_1} represents the relative risk corresponding to a unit change in Z_1 and e^{β^*} represents the quantity to which $e^{\hat{\beta}}$ converges. In this example omitting Z_2 from the model introduces a relatively small bias unless $|\alpha_2|$ is large. This again parallels the normal case mentioned at the start of this subsection.

Bretagnolle & Huber-Carol (1985) have also examined the effect on estimation of several parameters if covariates are omitted from the model. They have shown that when

Table 1. *Values of β^* when Z_2 is omitted from the model*

α_1	α_2	β^*	$\exp(\alpha_1)$	$\exp(\beta^*)$	α_1	α_2	β^*	$\exp(\alpha_1)$	$\exp(\beta^*)$
1·0	0·5	0·95	2·72	2·59	2·0	0·5	1·92	7·39	6·82
	1·0	0·85		2·34		1·0	1·73		5·64
	2·0	0·68		1·97		2·0	1·34		3·83
	3·0	0·59		1·80		3·0	1·10		3·00
1·5	0·5	1·43	4·48	4·18					
	1·0	1·28		3·60					
	2·0	1·01		2·75					
	3·0	0·86		2·36					

there is more than one parameter in the false model then there is a time up to which the sign of the bias of these parameters can be determined. This time depends on the distribution of the Z's and the values of the true parameters.

The results of this section indicate that the analysis based on the proportional hazards model will reflect the relative importance of the covariate when the true model is accelerated failure time, or proportional hazards with omitted covariates. In the latter case, $\hat{\beta}$ as an estimator of the regression parameter in the true model is asymptotically biased toward zero. The size of the bias will, however, be small unless $|\alpha_2|$ is large. A further analysis would consider evaluation of the asymptotic variance of $\hat{\beta}$ based on (2·4) when the true model is (2·1). These calculations are difficult, but in the case of missing covariates it seems likely that the asymptotic variance of $\hat{\beta}$ is less than the asymptotic variance of $\hat{\alpha}_1$ to complete the analogy with the normal case.

This work relates closely to a recent paper by Gail, Wieand & Piantadosi (1984). They considered the asymptotic bias in maximum likelihood estimators of regression parameters with omitted covariates in the exponential family and in parametric models with proportional hazards.

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